Example 8.12

Women, at child bearing age, of a district are grouped into seven age groups. The number of women lived during a calendar year in the district and the number of live births recorded during the same period are as follows:

Age Group (in years)	15-20	20-25	25-30	30-35	35-40	40-45	45-49
No. of Persons	20,000	22,000	28,000	32,000	29,000	24,000	8,000
No. of Live Births	50	1500	1700	2,000	1800	500	80

Calculate the general fertility rate of the district.

Solution:

The total number of women, at child bearing age in the district during the study period can be calculated from the given information as

$$\sum_{i=15}^{49} P_t^i = 1{,}63{,}000$$

and the total number of live births as

$B_t = 7{,}630.$

Hence, the general fertility rate of the district is

$$GFR = \frac{7630}{163000} \times 1000$$

$= 46.81$ *per thousand.*

(iii) Specific Fertility Rate (SFR)

It is a well known fact that fertility is affected by several factors such as age, marriage, migration, region *etc*. But, both *CBR* and *GFR* do not take into account of this fact. In this respect, **Specific Fertility Rate** (*SFR*) is defined as

$$SFR = \frac{\text{Number of live births to the women population in the reproductive age groups of specific section in a given period}}{\text{Total number of women in the reproductive age groups of the specific section in the given period}} \times 1000$$

SFR can be calculated separately for various age groups of females who are at child bearing age such as 15-20, 20-25, and so on. The *SFR* computed with respect to different reproductive age of women is known as the **Age Specific Fertility Rate** (*ASFR*), which can be calculated using the formula

$$ASFR(x, x+n) = \frac{B_t(x, x+n)}{P_t(x, x+n)} \times 1000$$

where

$B_t(x, x+n)$ denotes the number of live births to the women in the reproductive age group $(x,x+n)$ during the period t in the given region, and

$P_t(x, x+n)$ denotes the number of women in the reproductive age group $(x,x+n)$ during the period t in the given region.

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